

KOUVOLA ANJALA, Finland

WMO: 028300

Lat: 60.696N

Lon: 26.811E

Elev: 108

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.8	(c) -3.7	(d) -12.8	(e) 2.8	(f) -8.7	(g) -7.0	(h) 3.8	(i) -3.3	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	82.2	67.0	78.8	64.7	75.3	63.0	69.5	77.9	67.3	75.4	65.1	72.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	97.7	73.8	64.3	90.5	71.5	62.1	83.8	68.4	33.6	77.9	31.8	75.8	30.1	72.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -11.2	(f) 85.1	(g) 8.7	(h) 4.5	(i) -17.5	(j) 88.3	(k) -22.6	(l) 91.0	(m) -27.5	(n) 93.5	(o) -33.8	(p) 96.8		
(4) N/A	(5) N/A	(6) N/A	(e) -11.6	(f) 71.1	(g) 8.7	(h) 3.0	(i) -17.8	(j) 73.2	(k) -22.9	(l) 75.0	(m) -27.7	(n) 76.6	(o) -34.0	(p) 78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	19.7	23.1	28.9	39.8	51.7	58.6	64.2	60.9	52.7	41.2	34.8	27.5		
	DBStd	16.78	12.19	10.80	7.78	6.15	7.61	6.11	5.98	5.38	5.76	6.30	7.46	9.96		
	HDD50	4208	940	754	654	311	68	7	0	2	40	278	456	698		
	HDD65	8490	1405	1174	1119	755	415	208	90	148	369	738	906	1163		
	CDD50	1298	0	0	0	7	122	264	439	340	121	5	0	0		
	CDD65	106	0	0	0	0	4	15	65	22	0	0	0	0		
	CDH74	989	0	0	0	0	69	189	574	157	0	0	0	0		
	CDH80	192	0	0	0	0	8	25	131	28	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	25.6	2.0	1.7	1.5	1.2	1.6	2.4	2.4	2.8	2.5	2.7	2.6	2.3		
(15) Precipitation	PrecMax	35.7	3.6	4.5	2.8	2.9	4.1	4.6	4.8	5.6	5.8	5.9	5.6	6.5		
	PrecMin	17.8	0.4	0.1	0.1	0.3	0.4	0.4	0.3	0.2	0.6	0.7	0.3	0.5		
	PrecStd	4.0	0.9	1.0	0.7	0.6	1.0	1.0	1.1	1.3	1.4	1.3	1.1	1.1		
	PrecStd	4.0	0.9	1.0	0.7	0.6	1.0	1.0	1.1	1.3	1.4	1.3	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	41.9	49.4	68.3	80.7	82.9	87.4	83.5	70.4	57.3	50.7	46.8		
		MCWB	38.3	37.2	41.2	52.1	60.5	64.6	69.5	68.0	61.8	52.6	48.5	45.8		
	2%	DB	37.6	38.0	44.5	62.4	75.2	79.2	83.8	78.3	66.8	54.3	48.1	42.7		
		MCWB	36.9	35.8	38.2	48.7	58.7	63.1	68.9	65.5	59.7	51.4	47.2	41.5		
	5%	DB	35.9	36.1	40.6	56.3	71.3	75.6	80.3	74.3	64.3	52.8	45.7	40.4		
		MCWB	35.3	35.0	36.3	45.6	56.7	62.0	66.8	63.7	57.9	50.4	44.7	39.4		
	10%	DB	33.9	34.7	38.0	51.6	67.2	71.8	76.8	71.4	62.0	50.8	44.0	38.0		
		MCWB	33.5	33.9	34.9	42.9	54.2	59.5	64.9	61.8	57.3	48.9	43.3	37.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	37.8	42.0	52.7	64.2	68.5	72.4	71.6	63.6	54.6	49.1	46.3		
		MCDB	39.0	40.0	47.2	64.3	75.3	78.8	81.9	80.4	67.8	56.1	50.5	46.9		
	2%	WB	37.0	36.5	39.1	50.0	61.2	65.7	70.7	67.8	61.5	52.8	46.9	41.7		
		MCDB	37.5	37.7	42.7	59.8	70.4	74.6	80.2	74.7	64.7	53.8	47.8	42.4		
	5%	WB	35.5	35.2	37.3	46.8	58.5	62.9	68.8	65.4	59.9	51.0	44.9	39.4		
		MCDB	35.9	36.0	39.7	54.1	68.1	72.4	77.1	71.6	62.8	52.1	45.5	40.1		
	10%	WB	33.5	34.0	35.7	44.3	55.7	60.8	66.8	63.5	58.1	49.1	43.3	37.3		
		MCDB	33.9	34.7	37.5	50.0	65.0	69.5	74.7	68.8	60.9	50.4	43.9	37.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	9.7	13.2	16.9	20.2	18.0	17.7	16.6	13.7	10.6	6.8	7.4		
		MCDBR	6.3	7.8	15.4	26.3	26.4	24.6	22.7	21.4	17.9	12.7	8.0	7.9		
		MCWBR	6.0	6.8	10.2	14.6	12.5	11.4	9.8	10.6	11.2	9.6	7.7	7.9		
	5% WB	MCDBR	6.6	7.0	12.4	22.8	23.2	21.2	20.0	17.7	14.5	10.9	7.8	7.7		
		MCWBR	6.4	6.3	8.7	13.6	11.8	11.0	9.3	9.6	9.7	9.3	7.6	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.308	0.331	0.369	0.367	0.363	0.385	0.375	0.349	0.330	0.296	0.376		
	taud		2.120	2.169	2.230	2.311	2.405	2.451	2.409	2.431	2.488	2.393	2.197	2.592		
	Ebn at Noon		146	210	249	261	272	275	265	258	245	206	135	100		
	Edh at Noon		13	23	30	33	33	32	33	30	24	19	12	9		
(44) All-Sky Solar Radiation	RadAvg		78	272	744	1218	1660	1733	1729	1319	778	323	86	41		
	RadStd		17	58	91	130	181	111	159	140	91	49	15	5		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	-288	-310	N/A	N/A

Nomenclature: See separate page